

**Phonotactic conditions of geminates can influence sound symbolic effects:
Cuteness and evilness judgement experiments for Japanese speakers**

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Abstract

This study explores the sound symbolic effects of geminate consonants. I propose three working hypotheses: 1) geminates convey a particular meaning more strongly than singletons, 2) geminates convey it to the same extent as singletons, and 3) geminates convey it less strongly than singletons. To evaluate the most plausible hypothesis, I conducted experiments in which Japanese speakers judged cuteness and evilness of nonce words using a four-point scale. The results indicate that, on the one hand, the low-frequency geminate /rr/ in Japanese conveys cuteness less strongly than its singleton counterpart. On the other hand, the low-frequency geminate /bb/ conveys evilness to the same extent as its corresponding singleton. In conclusion, this study suggests that both the second and third hypotheses are plausible in the context of sound symbolism. Further research is needed to determine whether these findings extend to other sound symbolic images across languages.

1. Introduction

One of the most distinctive features of human languages is the arbitrariness of the relationship between linguistic sounds and meaning (Hockett 1963). This arbitrariness allows the phonological forms of words to be freely determined by language communities, as long as they do not violate the phonotactic constraints of each language. However, languages also exhibit motivated relationships between sound and meaning (Lockwood & Dingemanse 2015; Perniss et al. 2010). Specifically, phonological forms can encode systematic sound-meaning correspondences, a phenomenon generally referred to as sound symbolism (e.g. Hinton et al. 1994). A well-known example of sound symbolism is found in onomatopoeic words, also known as mimetic words or ideophones (Akita & Dingemanse 2019). For instance, in Japanese, the words *korokoro* and *gorogoro* both describe the motion of rolling objects. However, *korokoro*, which contains a voiceless /k/, evokes the image of a smaller, lighter object rolling, whereas *gorogoro*, which contains a voiced /g/, suggests a larger, heavier object rolling (Hamano 1986).

A growing body of research on sound symbolism (Nielsen & Dingemanse 2021) indicates that this phenomenon is pervasive both within the lexicon of a single language and across multiple languages (Sidhu 2024; Sidhu & Pexman 2019a, b). For example, corpus-based studies have identified phonological features that systematically recur in nouns and adjectives with related meanings (e.g. Blasi et al. 2016; Johansson et al. 2020; Joo 2020; Winter & Perlman 2021). Moreover, experimental studies using nonce words have demonstrated that sound-meaning mappings extend beyond attested vocabulary (e.g. Akita et al. 2024; Shinohara & Kawahara 2010), highlighting the productive nature of sound symbolism in language (Kawahara & Kumagai 2019a).

This study focuses on the sound-meaning effects of double consonants, or geminate consonants. Previous research on sound symbolism has paid little attention to geminates. One reason is that some studies assume that geminates hold no special status in sound symbolism and treat them the same way as their corresponding singletons or single consonants. For example, Uno et al. (2020), who analysed the names of ‘villainous’ characters in *Pokémon*, a well-known Japanese computer game, stated in their methodology section: ‘Geminate consonants (i.e. long consonants) were counted as one instance, not two’ (Uno et al. 2020:442).¹

Another reason for the limited focus on geminates in sound symbolism is that their behaviour varies across languages. Many languages, such as Arabic, Finnish, Italian, and Japanese, exhibit a phonemic contrast between singletons and their corresponding geminates (e.g. Ridouane 2010, for further examples). However, not all languages make this distinction. For instance, English, Spanish, and Hawaiian lack such a contrast. Even among languages that permit geminates, their distribution and phonemic status differ. In some languages, all singleton consonants can contrast with their geminate counterparts, whereas in others, only a subset of singletons exhibits this contrast (e.g. Blevins 2008; Maddieson 2008; Podesva 2002).

Additionally, the permissible positions of geminates vary cross-linguistically. While some languages allow geminates in word-initial, medial, or final positions, word-medial geminates are the most common (e.g. Pajak 2013; Thurgood 1993; see also Dmitrieva 2012). Other cross-linguistic differences concerning the phonological status of geminates

¹ Another study mentioning geminates is Kawahara et al. (2018), who showed in the Appendix that the presence of geminates was negatively correlated with the weight of *Pokémon* characters but provided no explanation for this correlation.

have been widely discussed in the phonological literature (e.g. Topintzi & Davis 2017). These variations across languages may complicate efforts to investigate how geminates function in sound-meaning correspondences.

This study introduces three working hypotheses regarding the sound symbolic effects of geminates, as outlined in (1). Suppose here that a single consonant (C) is associated with a meaning or image (I). The first hypothesis (1a) proposes that a geminate (CC) conveys this meaning more strongly ($I + \alpha$) than its singleton counterpart. This idea aligns with the cumulative effect of sound symbolism, which suggests that the presence of two identical sounds is more likely to amplify a particular meaning or image than a single occurrence (e.g. D’Onofrio 2014; Kawahara & Kumagai 2021; Thompson & Estes 2011). Additionally, this hypothesis is plausible because geminates sometimes function to intensify word meanings (e.g. Japanese: /kata-i/ ‘solid’ → [kattaai], /kusa-i/ ‘stink’ → [kussaai]). This is similar to reduplicated forms, in that ‘double’ forms can express intensity (Fischer 2011; Li & Ponsford 2018). If this hypothesis holds, geminates can be interpreted quite literally as ‘double’ consonants in sound symbolic grammar. The second hypothesis (1b) suggests that a geminate (CC) conveys the meaning or image to the same extent as a singleton, implying that in sound symbolism, it makes no difference whether the consonant associated with the meaning is a singleton or a geminate (e.g. Uno et al. 2020). Finally, the third hypothesis (1c) posits that a geminate (CC) conveys the meaning or image less strongly ($I - \alpha$) than the singleton (=1c). This attenuation may occur when sound symbolic effects are weakened by phonotactic constraints on geminates, such as markedness.

- (1) Three working hypotheses regarding the sound symbolic effects of geminates
- a. Geminates convey a particular meaning or image more strongly ($I + \alpha$) than singletons.
 - b. Geminates convey a particular meaning or image to the same extent as singletons.
 - c. Geminates convey a particular meaning or image less strongly ($I - \alpha$) than singletons.

The hypotheses (1a) and (1c) are contrastive, but both may coexist within a language or across languages. This is justified by the fact that natural languages exhibit morpho-phonological processes that may serve conflicting functions. For example, reduplication can express two similar concepts, such as intensification and iteration, but it also exhibits

contrastive functions, such as augmentation vs. diminution (Fischer 2011; Lǐ & Ponsford 2018). Therefore, two contrastive hypotheses can be simultaneously supported.

To test these hypotheses, this study conducted nonce word experiments with Japanese speakers, focusing on the sound symbolic effects of geminate consonants on perceptions of cuteness and evilness. The results showed that, on the one hand, a low-frequency geminate in Japanese conveys cuteness less strongly than its singleton counterpart, while on the other hand, another low-frequency geminate conveys evilness to the same extent as its corresponding singleton. Taking all the results into account, this paper argues that Hypotheses (1b) and (1c) remain plausible, whereas Hypothesis (1a) is not supported.

This study did not investigate sound symbolic trends in real words that express cuteness or evilness. One reason is that, as these abstract concepts are difficult to define, it was not easy to collect real words that suited the purpose of this research. Even if a statistical trend of sound symbolism does not appear in real words, it can still be observed in nonce word experiments or fictional names, and vice versa. This is because languages can generate sound symbolic associations independently of existing vocabulary. Another reason is that geminates are generally less frequent than singletons, making a fair comparison difficult. Geminates are more constrained by phonotactic restrictions than singletons (Section 2.1). Consequently, if a geminate appears less frequently than its corresponding singleton in words with a particular meaning, it is unclear whether this reflects its unsuitability for that meaning or whether phonotactic constraints are responsible for its suppression.

The current paper is structured as follows: Section 2 explains geminate consonants in Japanese, showing that /rr/ and /bb/ are low in frequency. Sections 3 and 4 examine whether the low-frequency geminates can influence a sound symbolic effect through nonce word experiments on cuteness and evilness, respectively. Section 5 discusses the results in relation to the three hypotheses. Section 6 provides concluding remarks for future research.

2. Background

2.1 Geminates in Japanese

Languages such as Arabic, Finnish, Italian, and Japanese exhibit a phonemic contrast between singleton and geminate consonants. In Japanese, for example, *kata* ‘shoulder’ and *katta* ‘bought’ are phonemically contrastive, as are *kaki* ‘oyster’ and *kakki* ‘vigour’.

In Japanese orthography, the first consonant of a geminate is known as *sokuon* (e.g. Kawagoe 2015; Kawahara 2015a) and is represented by a small letter <っ> or <ッ> (hiragana or katakana characters), as in /katta/ ‘bought’ <かっ た> and /kakki/ ‘vigour’ <かっ き> (cf. /kata/ ‘shoulder’ <か た> and /kaki/ ‘oyster’ <か き>).

Phonotactically, geminates occur word-medially in Japanese.² The first consonant of a geminate occupies the coda of the preceding syllable, while the second consonant forms the onset of the following syllable (e.g. /kat.ta/, /kak.ki/, where the period denotes a syllable boundary). From the perspective of moraic phonology (Hayes 1989; see also Davis 2011), geminates are inherently moraic (μ). As a result, words such as /ka^μt^μta^μ/ and /ka^μk^μki^μ/ are trimoraic, whereas /ka^μta^μ/ and /ka^μki^μ/ are bimoraic.

In general, geminate consonants are subject to certain phonotactic restrictions (e.g. Ito & Mester 1995, 1999; Ito et al. 2017; Kawahara 2015b). In Japanese, voiceless obstruents (e.g. /p, t, k, s/) can occur as geminates across different lexical strata (native words, Sino-Japanese (SJ) words, and loanwords).³ However, geminates of voiced obstruents, such as /bb, dd, gg, zz/, and those of the flap (/ɾ/) are restricted to loanwords.⁴

The present experiments (Sections 3 and 4) focused on /p/, /b/, /g/, and /ɾ/. Geminate /pp/ appears in native and SJ words, as well as in loanwords, as illustrated in (2a). This is because /h/ in the second member of compounds is replaced with [pp] in native and SJ words (e.g. Ito et al. 2017; Labrune 2012). In contrast, as shown in (2b), voiced geminates such as /bb, dd, gg/ occur exclusively in loanwords (e.g. Ito & Mester 1995, 1999). The frequency of voiced plosive geminates varies to some extent. Several dictionary-based studies have shown that /bb/ is less frequent than /dd/ and /gg/ (e.g. Hirayama 2008; Kawagoe & Arai 2002; Shirai 1999; see also Ito et al. 2017); that is, there are fewer examples with /bb/ than those with /dd, gg/ among loanwords whose source words end in /b, d, g/. According to Hirayama’s (2008) survey, there were only six examples with /bb/

² Word-final geminates are not found, and word-initial geminates appear only when a word or phrase is partially elided in casual speech (Kawahara 2015a, e.g. /urusai/ ‘shut up!’ → [ssai]; /toiuka/ ~ /teiuka/ rather; perhaps I should say ...’ → [tteka]).

³ Note that the geminate /h/ is very infrequent in native and SJ words (Labrune 2012:70), although it serves as a marker indicating that names containing it are loanwords from German (e.g. *mahha* ‘Mach’, *bahha* ‘Bach’; Kawahara & Kumagai 2024).

⁴ In addition, the glides /w, j/ are disallowed from geminating (Kawahara & Pangilinan 2017). It should be noted, however, that voiced obstruents and liquid undergo gemination to express emphasis (e.g. /jaba-i/ ‘exciting’ → [jabbaai], /karu-i/ ‘light’ → [karuui]; Kawahara 2015b). This also applies to glides /w, j/ (e.g. /kawai/ ‘scary’ → [kowwaai]).

out of 27 loanwords with final /b/ in the source words (e.g. *mobbu* ‘mob’, *nobbu* ‘knob’), whereas there were 35 examples with /dd/ out of 41 loanwords with final /d/, and 19 examples with /gg/ out of 36 loanwords with final /g/ (e.g. *kiddo* ‘kid’, *heddo* ‘head’, *reddo* ‘red’, *roddo* ‘rod’, *wiggu* ‘wig’, *eggu* ‘egg’, *foggu* ‘fog’, *reggu* ‘leg’). Additionally, as indicated in (2c), /rr/ is restricted to loanwords from Arabic or Italian (e.g. Kawahara & Kumagai 2024; Tanaka 2017). It is therefore likely to be less frequent than /dd/ and /gg/, which are more commonly found in loanwords from English. To summarise, /bb/, /rr/ are relatively infrequent in Japanese.

(2) a. Geminate /pp/ (high-frequency)

- Loanwords: /happii/ ‘happy’; /guppii/ ‘guppy’; /zippaa/ ‘zipper’
 Native: /suki/ ‘empty’ + /hara/ ‘stomach’ → /suki[pp]ara/ ‘empty stomach’
 /dasi/ ‘to emit (conjunctive)’ + /hanasi/ ‘to release (nominative)’ →
 /dasi[pp]anasi/ ‘to keep emitting’
 SJ: /situ/ ‘lose’ (失) + /hai/ ‘lose’ (敗) → /si[pp]ai/ ‘failure’
 /zetu/ ‘extinct’ (絶) + /han/ ‘version’ (版) → /ze[pp]an/ ‘out of print’

b. Geminate /bb/ (low-frequency) < /dd, gg/

- /bb/: /hobbuzu/ ‘Hobbes’ (English philosopher)
 /dd, gg/: /reddo/ ‘red’; /heddo/ ‘head’; /beddo/ ‘bed’;
 /eggu/ ‘egg’; /doggu/ ‘dog’; /baggu/ ‘bag’

c. Geminate /rr/ (low-frequency)

- /rr/: /arraa/ ‘Allah’ (God, from Arabic);
 /roberri/ ‘Rovelli’ (physician, from Italian);
 /ɸaruɸarre/ ‘farfalle’; /ɸeraari/ ~ /ɸerraari/ ‘Ferrari’

2.2 Motivation for experiments

The previous section established that /bb/ and /rr/ are relatively infrequent in Japanese. This observation leads to the prediction that Hypothesis (1c) is plausible; specifically, the phonotactic constraint that renders certain geminates infrequent may weaken the sound symbolic effects typically associated with their corresponding singletons.

Support for this prediction comes from research on cross-modal correspondences between sounds and shapes, particularly the maluma-takete or bouba-kiki effect (Köhler 1929/1947; Ramachandran & Hubbard 2001). Studies have shown that this effect fails

when auditory stimuli violate the phonological structure of a target language (Styles & Gawne 2017). For example, when Syuba speakers, a Central Bodish language of the Tibeto-Burman family, were presented with the stimuli /k^hik^hi/ and /bubu/, the bouba-kiki effect did not emerge. This failure was attributed to the fact that Syuba does not permit words ending in /u/, such as ‘bubu’. These findings suggest that phonotactic constraints can override or diminish sound symbolic associations. Following this reasoning, the next two sections investigate whether, as predicted by Hypothesis (1c), the low-frequency geminates in Japanese attenuate the sound symbolic effects of cuteness (Section 3) and evilness (Section 4).

3. Experiment 1: cuteness

3.1 Research questions

Experiment 1 focused on the concept of cuteness, or *kawaii* in Japanese. Investigating such an abstract quality presents challenges, particularly because, unlike size-related sound symbolism (Sapir 1929), there is no clear articulatory or acoustic basis for sound-meaning correspondences. Additionally, cultural differences influence how abstract qualities are perceived, making cross-linguistic comparisons difficult.

Despite these challenges, recent studies have begun to explore the phonological correlates of cuteness. Kumagai (2022) examined which phonological features are associated with cuteness in Japanese. The findings indicate that labial obstruents are perceived as ‘cuter’ in terms of place of articulation, sonorants (particularly /r/) in terms of manner of articulation, and voiceless obstruents in terms of voicing. These features are linked to characteristics such as babyishness, smallness, softness, roundness, and femininity, all of which contribute to perceptions of cuteness. Phonological features associated with cuteness have also been examined cross-linguistically through nonce word experiments involving speakers of German, Chinese, and Korean (Jang 2025; Kumagai & Moon 2021; Schmitz et al. 2023).

Building on this line of inquiry, Experiment 1 investigates whether geminate consonants contribute to sound symbolic effects associated with cuteness. In this study, the term *cuteness effect* refers to any effect that enhances the perceived cuteness of a name. This experiment examined /p/ and /t/, both hypothesised to contribute to a cuteness effect, where names containing /p/ or /t/ are perceived as cute by Japanese speakers. Previous research has indicated that /p/ is considered cuter than other consonants (Kumagai 2019), while /t/ received higher cuteness ratings than voiced coronal obstruents (Kumagai 2022).

As discussed in Section 2.1, /rɾ/ is less frequent in Japanese than /pp/. This experiment sought to determine whether these geminates exhibit different behaviours in terms of their sound symbolic effects.

3.2 Task and stimuli

Experiment 1 presented nonce words containing target singletons or geminates to Japanese speakers, who were asked to rate how cute they sounded using a four-point scale. Table 1 lists the stimuli used in this experiment. Four conditions were tested: /p/, /pp/, /ɾ/, and /rɾ/, with each condition comprising 10 items.

All stimuli were four morae in length (e.g. ta^u:^upa^uN^u, ta^up^upa^uN^u). The target singletons or geminates appeared word-medially: (i) singleton [CV:_VN] (e.g. *taapan*); (ii) geminate [CV__VN] (e.g. *tappan*). A long vowel (:) was inserted before the target singleton in each singleton condition to maintain consistent length across conditions, based on the assumption that vowel length does not influence sound symbolic judgements made by Japanese speakers. To ensure consistency, non-target consonants (C) and vowels (V) were kept uniform across all conditions.

Table 1. Stimuli (Experiment 1)

p				ɾ			
singleton		geminates		singleton		geminates (infrequent)	
CV:pVN	(Jpn)	CVppVN	(Jpn)	CV:rVN	(Jpn)	CVrɾVN	(Jpn)
ta:paN	ターパン	tappaN	タッパン	ta:raN	ターラン	tarraN	タッラン
na:paN	ナーパン	nappaN	ナッパン	na:raN	ナーラン	narraN	ナッラン
ka:piN	カーピン	kappiN	カッピン	ka:riN	カーリン	karrin	カッリン
so:piN	ソーピン	soppiN	ソッピン	so:riN	ソーリン	sorrin	ソッリン
ke:puN	ケーブン	keppuN	ケッブン	ke:ruN	ケールン	kerrun	ケッルン
he:puN	ヘーブン	heppuN	ヘッブン	he:ruN	ヘールン	herrun	ヘッルン
yo:peN	ヨーペン	yoppeN	ヨッペン	yo:reN	ヨーレン	yorren	ヨッレン
tei:peN	チーペン	teippeN	チッペン	tei:reN	チーレン	teirren	チッレン
mu:poN	ムーボン	muppoN	ムッボン	mu:roN	ムーロン	murron	ムッロン
wa:poN	ワーボン	wappoN	ワッボン	wa:roN	ワーロン	warron	ワッロン

3.3 Procedure

The experiment was conducted in December 2024 using the online questionnaire platform SurveyMonkey. Participants were recruited through the platform's Buy Response Function, with Japan set as the target country.

Participants first signed a consent form and were eligible to proceed if they were native Japanese speakers who had never studied sound symbolism. They were then presented with the following instructions: 'Non-existing names are presented per question. Please judge how cute they are using a 4-point scale (4: very cute, 3: cute, 2: not very cute, 1: not cute).' Before the trial began, participants completed a practice question by rating the imaginary name 'ramissu' for cuteness. The average rating for this item was 2.5.

The names were presented in written form using katakana characters (Jpn in Table 1). In Japanese orthography, the first consonant of a geminate is represented by a small letter <ッ>. As a result, the letter denoting the target singleton or geminate (e.g. <パ> = /p/, /pp/) appeared only once in singleton items <タ^ッパン> (ta:paN) and geminate items <タ^ッッパン> (tappaN). During the trial session, participants answered a total of 40 questions, with the order randomised for each participant.

3.4 Participants

A total of 151 native Japanese speakers participated in Experiment 1. They were asked to select one option each from the gender options and age range options provided by the SurveyMonkey. The gender breakdown was as follows: Females = 72, Males = 79; age breakdown: 16 (18-29 years old), 43 (30-44 years old), 60 (45-60 years old), 32 (60+ years old).

3.5 Statistics

The results were analysed using Bayesian cumulative regression with the logit link function (Bürkner & Vuorre 2019; Gambarota & Altoè 2024; Tutz 2022), as the cuteness rating was measured on an ordinal scale. The dependent variable was the score (1~4), and the independent variables were consonant (*p* vs. *r*), the structure (*singleton* vs. *geminate*), and the interaction between the two. Non-target consonant and vowel quality were not included as fixed factors in the model. Random factors included by-participant and by-stimulus varying intercepts, and by-participant varying slopes.

The analysis was implemented using the *brms* package (Bayesian Regression Models using Stan, Bürkner 2017) in R version 4.2.2 (R Core Team 2022). The model ran four chains with 3,000 iterations each (the first 1,000 iterations per chain were warmups), yielding a total of 8,000 posterior samples. Model convergence was confirmed, as the $\hat{R}$ values for each parameter were 1.00. The files are available in the Open Science Framework (OSF) repository (<https://osf.io/7ph3y/>).

3.6 Results

Figure 1 presents a stacked bar chart showing the distribution of responses in Experiment 1. Each response on the 4-point scale was distinguished by using different colours (from left to right: ‘1: not cute’ to ‘4: very cute’), and the percentage for each score was indicated. The mean scores for each condition were as follows: $/p/=1.79$, $/pp/=1.75$, $/r/=1.85$, and $/rr/=1.63$.

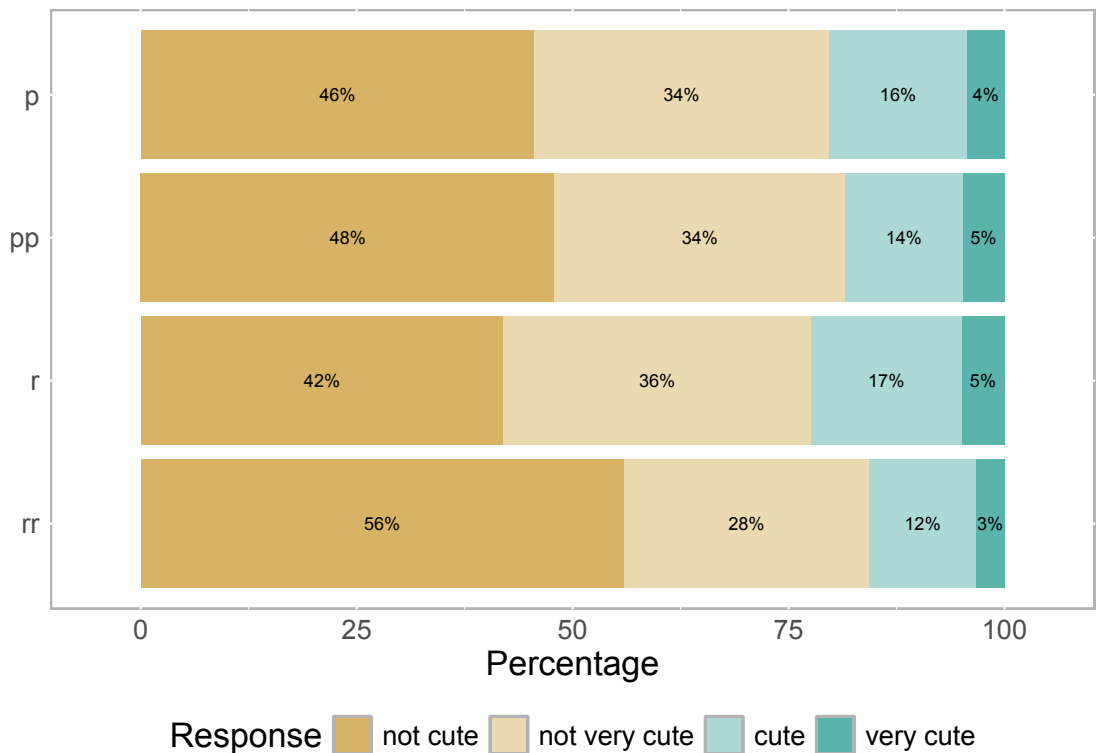


Figure 1. Cuteness judgement results for Experiment 1

Table 2 displays the coefficient (β), error, and 95% Credible Interval (CI) for each parameter (note that intercepts are excluded). A credible effect was assumed if the 95% CI of an estimated parameter did not include 0. In this analysis, *p* and *singleton* served as

the reference levels. The comparison of *r* (vs. *p*) yielded a positive coefficient ($\beta=0.29$), but its 95% CI included 0 ($[-0.15, 0.74]$), indicating no credible difference between the two consonants. Comparing geminates with singletons also revealed no credible difference ($\beta=-0.32$; 95% CI= $[-0.78, 0.14]$). However, the interaction between *r* and *geminate* produced a negative coefficient ($\beta=-1.07$), and its 95% CI did not include 0 ($[-1.73, -0.43]$). Therefore, the cuteness rating decreased when *r* appears as a geminate.

Table 2. Statistical results for Experiment 1

	Coefficient (β)	Error	95% CI
<i>r</i> (vs. <i>p</i>)	0.29	0.22	$[-0.15, 0.74]$
<i>geminate</i> (vs. <i>singleton</i>)	-0.32	0.23	$[-0.78, 0.14]$
<i>r:geminate</i>	-1.07	0.33	$[-1.73, -0.43]$

3.7 Discussion

To summarise the results, no difference was found in cuteness between the consonants (*r* vs. *p*), nor between structure (*singleton* vs. *geminate*). However, the coefficient for the interaction between *r* and *geminate* was statistically skewed towards negative values. This suggests that *r* diminished the cuteness effect when it appeared a geminate, whereas *p* did not. A possible explanation for this result is that /rr/ is infrequent in Japanese, which may have attenuated the cuteness effect associated with *r*. The finding that phonological violations influence sound symbolic judgements aligns with the argument presented in Styles & Gawne (2017), as discussed in Section 2.2.

4. Experiment 2: evilness

4.1 Hypothesis tested

Evilness is another semantic aspect explored in this study. Like cuteness, it is an abstract concept, making it difficult to pinpoint direct motivations for the sound-meaning correspondence. However, recent studies have identified particular phonological features associated with evilness. A survey of *Disney* and *Pokémon* character names has shown that villains and sinister-looking characters (e.g. *Gideon* in *Pinocchio*; *Jafar* in *Aladdin*) are more likely to have voiced obstruents in their names than voiceless obstruents (Uno et al. 2020).

Subsequent research has also demonstrated that voiced obstruents are perceived as more appropriate for villainous character names by speakers of Japanese, English, and Brazilian Portuguese (e.g. Godoy et al. 2021; Kawahara & Kumagai 2019b; Kawahara et

al. 2021). These studies argue that the aerodynamic difficulties involved in producing voiced obstruents can be linked to negative perception, specifically evilness. In this study, the term *evilness effect* refers to an influence that makes names sound evil or villainous.

Experiment 2 focused on /b/ and /g/. It is assumed that both /b/ and /g/ are *evil* consonants, and that names containing them are perceived as evil by Japanese speakers. As discussed in Section 2.1, /bb/ is less frequent than /gg/ (e.g. Hirayama 2008). Experiment 2 examined whether /bb/ would more likely diminish its evilness effect than /gg/.

4.2 Task and stimuli

The task was identical to that of Experiment 1. The stimuli are shown in Table 3. Experiment 2 examined four conditions: /b/, /bb/, /g/, and /gg/. The number of items per condition and the mora length of each item matched those in Experiment 1. Voiced obstruents and [p] were excluded as initial consonants (C), as they can trigger the devoicing of voiced obstruent geminates (Fukazawa et al. 2015; Kawahara 2006; Nishimura 2006).

Table 3. Stimuli (Experiment 2)

b				g			
singleton		geminates (infrequent)		singleton		geminates	
CV:bV _N	(Jpn)	CVbbV _N	(Jpn)	CV:gV _N	(Jpn)	CVggV _N	(Jpn)
ra:ba _N	ラーバン	rabba _N	ラッバン	ra:ga _N	ラーガン	ragga _N	ラッガン
na:ba _N	ナーバン	nabba _N	ナッバン	na:ga _N	ナーガン	nagga _N	ナッガン
ka:bi _N	カービン	kabbi _N	カッビン	ka:gi _N	カーギン	kaggi _N	カッギン
so:bi _N	ソービン	sobbi _N	ソッビン	so:gi _N	ソーギン	soggi _N	ソッギン
ke:bu _N	ケーブン	kebbu _N	ケッブン	ke:gu _N	ケーグン	keggun _N	ケッグン
hi:bu _N	ヒーブン	hibbu _N	ヒッブン	hi:gu _N	ヒーグン	higgun _N	ヒッグン
yo:be _N	ヨーベン	yobbe _N	ヨッベン	yo:ge _N	ヨーゲン	yogge _N	ヨッゲン
ru:be _N	ルーベン	rubbe _N	ルッベン	ru:ge _N	ルーゲン	rugge _N	ルッゲン
mu:bo _N	ムーボン	mubbo _N	ムッボン	mu:go _N	ムーゴン	muggo _N	ムッゴン
ne:bo _N	ネーボン	nebbu _N	ネッボン	ne:go _N	ネーゴン	neggo _N	ネッゴン

4.3 Procedure

The experiment was conducted online in March 2025 using the SurveyMonkey platform. As in Experiment 1, participants were recruited via the Buy Response Function. They were asked to rate how appropriate each name was for a villainous character using a 4-point scale (4: very appropriate, 3: appropriate, 2: not very appropriate, 1: not appropriate). A practice question was included in which participants rated the imaginary name 'ramisshu' for villainousness; the average rating was 2.09. All other procedures were identical to those in Experiment 1.

4.4 Participants

A total of 150 Japanese speakers participated in Experiment 2 (gender breakdown: Females = 54, Males = 96; age breakdown: 9 (18-29 years old), 47 (30-44 years old), 60 (45-60 years old), 34 (60+ years old)).

4.5 Statistics

The statistical analysis followed the same procedure as in Experiment 1, employing Bayesian ordinal logistic regression. The only difference in the present analysis was the coding of the consonant (*b* vs. *g*). The model ran four chains with 4,000 iterations each (the first 1,000 iterations per chain were warmups), generating a total of 12,000 posterior samples. The $\hat{R}$ values for all parameters confirmed successful convergence at 1.00. The files are available in the OSF repository.

4.6 Results

Figure 2 presents a stacked bar graph showing the results of Experiment 2. The average scores for each condition were as follows: /b/ = 2.17, /bb/ = 2.12, /g/ = 2.33, and /gg/ = 2.24.

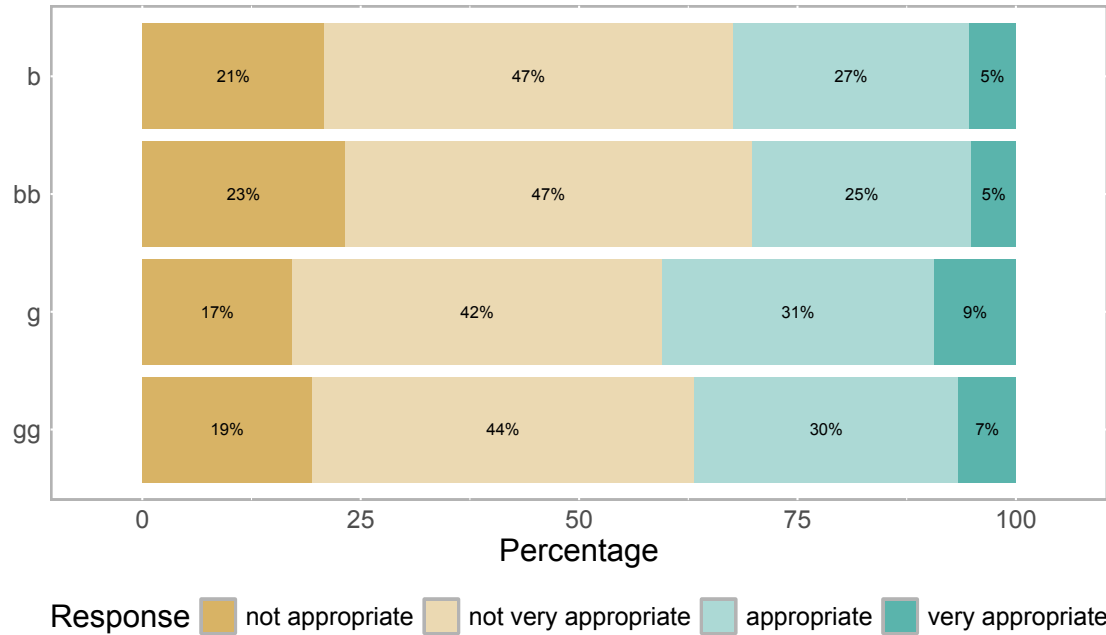


Figure 2. Evilness judgement results for Experiment 2

Table 4 presents the statistical results, with *g* and the *singleton* structure serving as the reference levels. The coefficient for *b* was negative ($\beta = -0.49$), and its CI did not include 0 ($[-0.74, -0.25]$), suggesting that the evilness rating for *b* was lower than that for *g*. A comparison between the two structures also showed a negative coefficient ($\beta = -0.31$), and its CI did not include 0 ($[-0.58, -0.04]$), indicating that geminates were rated lower than singletons. However, the coefficient for the interaction between *b* and *geminate* was 0.10, with a CI that included 0 ($[-0.22, 0.44]$). Therefore, *g* appears to diminish its evilness effect when geminated, whereas *b* does not.

Table 4. Statistical results for Experiment 2

	Coefficient (β)	Error	95% CI
<i>b</i> (vs. <i>g</i>)	-0.49	0.12	$[-0.74, -0.25]$
<i>geminate</i> (vs. <i>singleton</i>)	-0.31	0.14	$[-0.58, -0.04]$
<i>b:geminate</i>	0.10	0.17	$[-0.22, 0.44]$

4.7 Discussion

Although it was initially assumed that both *b* and *g* would equally convey an image of evilness, the comparison between the singletons suggests that *g* was more strongly associated with evilness than *b*. This finding has not been explicitly reported in the

experimental studies cited in Section 4.1. However, as this is not the primary focus of the present study, further investigation into this phenomenon is left for future research.

The mean rating for geminates was lower than that for singletons, but this result may be driven by the behaviour of /gg/, as there was no credible effect observed for /bb/. The important finding here is that *b* did not lose its evilness effect even when geminated. This contrasts with the results of Experiment 1, where /rr/ was rated lower in cuteness than /r/. This discrepancy will be discussed further in Section 5.

5. Overall discussion

We predicted that, since low-frequency geminates are marked, they could influence sound symbolic effects related to cuteness and evilness. To summarise the experimental results, /rr/ received lower cuteness scores than /r/, whereas the score for /bb/ was almost the same as that for /b/. We predicted the results for /rr/ based on the previous finding of Styles & Gawne (2017), in which the bouba-kiki effect failed due to a phonotactic violation in the stimuli. However, the results for /bb/ was unexpected. Therefore, it can be concluded that a consonant may diminish a sound symbolic effect when it becomes a low-frequency geminate, but not always.

This raises the question of why there is a discrepancy between the effects on cuteness and evilness. One possible explanation lies in the nature of sound symbolism; some sound symbolic associations may be stronger than others. It has been argued that sound symbolic phenomena are stochastic rather than deterministic (Alderete & Kochetov 2017; Dingemanse 2018; Kawahara 2020; Sidhu 2024), suggesting that certain associations are more likely to manifest than others. The present experiments showed that the evilness scores of /b, g/ in Experiment 2 exceeded 2.0, whereas the cuteness scores of /p, r/ in Experiment 1 ranged between 1.79 and 1.85. These results suggest that the association between voiced obstruents and evilness may be stronger than that between /p, r/ and cuteness, at least in Japanese. This could explain why /bb/ did not diminish the evilness effect.

Based on these experimental results, we now discuss which hypothesis presented in Section 1 (repeated in (3)) is most plausible. The results of Experiment 1 indicated that /rr/ conveys cuteness less strongly than /r/, supporting Hypothesis (3c). In contrast, the results of Experiment 2 showed that /bb/ conveys evilness to the same extent as /b/, which may support Hypothesis (3b). There was no case in which geminates received higher

scores than singletons in either experiment. Considering all these findings, this paper argues that Hypotheses (3b) and (3c) remain possible explanations.

(3) Three hypotheses regarding the sound symbolic effects of geminate consonants (= (1))

Geminates convey a meaning or image ...

- a. ... more strongly ($I + \alpha$) than singletons.
- b. ... to the same extent as singletons. → Possible.
- c. ... less strongly ($I - \alpha$) as singletons. → Possible.

We then discuss how the current results can be analysed from the perspective of theoretical phonology. Phonological forms involving sound symbolic effects can violate the phonotactic constraints of a language. For example, expressive palatalisation in Japanese babytalk allows the production of [ɕɕ] (e.g. /sensei/ ‘teacher’ → [ɕensei]), a sequence otherwise permitted only in loanwords (Alderete & Kochetov 2017). Similarly, the word *hime* ‘princess’ alternates with *pime* in the Japanese nicknaming process (*pime-yobi*) but does not undergo *rendaku* to become *bime*, despite both forms violating the OCP-labial constraint (e.g. *Ayu* (name) + *hime* ‘princess’ → *Ayu-pime* ‘Princess Ayu’; *mai* ‘dancing’ + *hime* → *mai-hime* ‘dancing girl’; **mai-bime*; Kumagai 2023). The results of Experiment 2 suggest that the evilness effect is not influenced by the phonotactic restriction against [bb] (i.e. *bb) within the evilness-expressive grammar, a subgrammar that applies when speakers judge evilness. In contrast, the results of Experiment 1 indicate that sound symbolism is sometimes susceptible to the phonotactic constraints of a language. Within the cuteness-expressive grammar, which governs judgements of cuteness, the perception of cuteness appears to be suppressed by the markedness constraint that prohibits [rr] (i.e. *rr). Further research should clarify which specific phonotactic constraints are enforced in different expressive subgrammars.

6. Concluding remarks

Phonotactics vary from language to language. Therefore, even if two or more languages share a particular sound-meaning association, the strength of the effect may differ across languages. In this respect, phonotactic constraints can override or diminish sound symbolic effects. Following Styles & Gawne (2017), this study has demonstrated that sound symbolic effects can be inhibited by phonotactics, thereby concluding that a consonant can diminish a sound symbolic effect when it becomes a geminate.

Neither experiment in this study provided evidence to support Hypothesis (1a=3a). This suggests that geminates are not perceived as ‘double’ consonants in phonology but rather as a single unit. Future research should explore whether this prediction holds across other sound symbolic studies in a variety of languages. It may be valuable to conduct follow-up experiments in languages that contrast singletons and geminates in word-initial positions. For example, Leti (Austronesian language) contrasts /r/ and /rr/ (e.g. *rusa* ‘to nail’ vs. *rrusa* ‘they nail’; Hume et al. 1997; Muller 2001). Also, Cypriot Greek features a minimal pair involving /p/ and /pp/, such as *pefti* ‘Thursday’ vs. *ppefti* ‘fall (3rd. sig.)’ (Christodoulou 2007; Muller 2001). Word-initial consonants may exhibit stronger sound symbolic images than their non-word-initial counterparts, as they are psychologically more prominent (Kawahara et al. 2008; Shinohara & Uno 2022). Therefore, word-initial geminates may convey a meaning or image more strongly than non-word-initial singletons, even if the former are less frequent. This may offer potential support for Hypothesis (3a).

Furthermore, as the present study has shown that geminate effects exhibited differing degrees between cuteness and evilness, it would be valuable to investigate whether this prediction varies across tested meanings or images. Given the limited research on the sound symbolic effects of geminates, it is expected that these questions will be addressed in future studies.

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